



Role of integrin and its potential as a novel postmortem biomarker in traumatic axonal injury

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Abstract

Traumatic axonal injury (TAI) accounts for a large proportion of the mortality of traumatic brain injury (TBI). The diagnosis of TAI is currently of limited use for medicolegal purposes. It is known that axons in TAI are diffusely damaged by secondary processes other than direct head injury. However, the physiopathological mechanism of TAI is still elusive. The present study used RGD peptide, an antagonist of the mechanotransduction protein integrin, to explore the role of integrin-transmitted mechanical signalling in the pathogenesis of rat TAI. The rats were subjected to a linearly accelerating load, and changes in beta-amyloid precursor protein (β -APP) expression, skeleton ultrastructure, skeleton protein neurofilament light (NF-L), and α -tubulin in the brainstem were observed, indicating that RGD could relieve the severity of axonal injury in TAI rats. In addition, the expression of β -integrin was stronger and centralized in the brainstem of the deceased died from TAI compared to other nonviolent causes. This study examined the pathophysiology and biomechanics of TAI and assessed the role of integrin in the injury of microtubules and neurofilaments in TAI. Thus, we propose that integrin-mediated cytoskeletal injury plays an important role in TAI and that integrin has the potential as a biomarker for TAI.

Keywords Traumatic axonal injury · Brainstem · Integrin · Microtubule · Neurofilament

Introduction

Traumatic brain injury (TBI) is a leading cause of violent death worldwide [1] and constitutes a considerable proportion of cases requiring forensic examination. Traumatic

axonal injury (TAI) refers to extensive axonal injury caused by mechanical violence of head trauma and accounts for a large proportion of TBI-induced deaths and severe disability [2–4]. The clinicopathological entity of TAI has evolved over the last 30 years. The typical characteristics of TAI are axonal swelling, axotomy, and even axonal bulbs. However, histopathology only plays a role in postmortem diagnosis but not in a clinical setting. Thus, the studies on TAI are significant not only in the clinical context but also in the forensic field.

It is generally accepted that primary axotomy, which occurs at the moment of trauma, is a rare form of axon damage and is only seen in massive TBI [5, 6]. A lot of evidence has shown that axonotmesis, which is also referred to as secondary axotomy, is mainly caused by a cascade of biochemical pathways in response to an applied mechanical force [7–10]. For forensic pathologists, the lack of a well-documented connection between assault and neuropathological findings in TAI means that the interpretation of cases is challenging. Thus, understanding the mechanism of secondary axotomy is needed for medicolegal purposes.

Integrin is a mechanosensitive protein that plays an important role in cellular mechanics [11–13]. Hemphill et al. showed that force transmission via integrin-binding proteins

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produced widespread neuron and axon injury [14]. On the other hand, an increasing number of studies have indicated that the intracellular cytoskeleton is pivotal in the cellular biomechanical mechanism [15, 16]. Two primary constituents of the cytoskeleton, microtubules, and neurofilaments, are structurally and functionally important mechanical components that not only provide a physical network structure for axons but also provide the “track” for transportation and the attachment points for many signalling proteins. The breakage and depolymerization of microtubules in TAI cause the interruption of axoplasmic transport, which leads to the local accumulation of protein and organelles and subsequent axonal swelling and retraction [17, 18]. One typical protein involved in this process, beta-amyloid precursor protein (β -APP protein), which is usually transported along axons to the synapse, accumulates to detectable levels due to transport impairment. Currently, β -APP immunostaining is regarded as an important indicator for the pathological diagnosis of TAI [5, 19]. Furthermore, it has been demonstrated that integrin-mediated mechanotransduction may regulate the rigidity and viscosity of the cytoskeleton [20, 21]. Thus, we hypothesized that the integrin-mediated mechanotransduction signalling pathway plays a critical role in the secondary axotomy of TAI by affecting the skeletons of axons.

In the present study, to explore the role of integrin in TAI and its effect on the skeleton of axons in TAI, we utilized RGD peptide, a kind of blocker of integrin, to block the integrin-mediated signalling pathway in TAI model rats. Then, we observed changes in the protein levels of microtubules and neurofilaments and the ultrastructure of the skeleton in the brainstem at three-time points postinjury in the TAI group, RGD group, and control group.

Material and methods

Animal model

All procedures were approved and monitored by the Animal Ethics Committee of Huazhong University of Science and Technology. Adult male Sprague–Dawley rats weighing 200–250 g (supplied by the Laboratory Animal Centre of Tongji Medical College) were used for the study. The TAI model was made by linear acceleration load referring to Marmarou method according to a previous report [22]. The rats were anesthetized by intraperitoneal injection of 0.4% pentobarbital sodium (30 mg/kg). Following a midline incision between coronal and lambdoid sutures, a steel disk 10 mm in diameter and 3 mm in thickness was adhered to the skull between the bregma and lambda sutures. Then, the animal was placed on a small sponge cushion in the prone position and fastened by a belt around the trunk, with the disk exactly at the center of the lower end of a 1 m plexiglass

tube. A cylindrical iron bar, 450 g in weight, was allowed to fall through the tube from a height of 1 m to impact the disk. The rat was moved at once when the bar hit the disk to prevent repeat impacts to the head.

Lateral ventricle injection

The vital signs of TAI rats were observed after the impact, and rats were quickly fixed to a stereotaxic apparatus after stabilization. Lateral ventricle puncture landmarks were located 1.5 mm from the rear side of the anterior fontanelle and 1.0 mm from the right side of the midline, and a hand-held craniotomy drill was used to make holes in the cranium. Then, RGD PBS solution (10 μ l, 1 nmol/ μ l) was injected through the hole into the lateral ventricles by a sterilized microsyringe according to a previous report [23].

The experimental group and sample processing

Seventy rats were randomly divided into the control group, TAI group, and RGD group (RGD was administered to TAI rats). The rats in the control group underwent the surgical procedure but were not subjected to cranial trauma. The rats in the RGD group were subjected to a blow and then followed by an intracerebroventricular injection of RGD (Sigma–Aldrich, A8052, USA). The rats in the TAI group were only injected with PBS solution. The rats in the TAI group and RGD group were sacrificed at 12 h, 24 h, and 72 h postinjury. The brain was removed and hemisected along the midline, half of the brainstem was stored in vials at -80°C for western blotting, and the other half of the brainstem was fixed in 4% paraformaldehyde for 48 h for immunohistochemical assessment.

Transmission electron microscopy (TEM)

The rats ($n=3$) were deeply anesthetized with 10% chloral hydrate and then perfused through the heart with pre-cooled heparinized saline (1000 units/l) and 2.5% glutaraldehyde + 4% paraformaldehyde at 24 h postinjury for TEM investigation. Then, the brains were removed, and the cerebral peduncle was trimmed into blocks of 1 mm $\times$ 1 mm and routinely fixed, rinsed, dehydrated, and embedded on the sagittal plane in epoxy resin Epon812. Ultrathin sections were prepared, stained with uranyl acetate and lead citrate, and examined with a Philips EM208s transmission electron microscope.

Immunohistochemistry of β -APP expression

The fixed brainstem tissue (from rostral to caudal) in paraformaldehyde was dehydrated with a gradient of sucrose solutions and then embedded in the optimal cutting temperature (OCT) compound on the sagittal plane. The blocks were cut into 14- μ m thick cryosections and dried at room temperature

for immunohistochemical staining. Routine antigen retrieval was performed in 0.1 M sodium citrate (pH=6.5) at 100 °C for 10 min. The sections were incubated with β -APP monoclonal antibody (1:5000, Abcam, ab105122, USA) at 4 °C overnight, followed by biotinylated secondary antibody, SABC reagent (Boster, Wuhan, China), and DAB solution at 37 °C for 30 min. The sections were dehydrated in graded alcohols, cleared in xylene, and covered.

Western blotting

The – 80 °C-stored hemi-brainstem was homogenized, resolved by SDS–PAGE, and transferred to a PVDF membrane using a Mini-PROTEAN Tetra Cell (Bio-Rad). The membranes were blocked with 5% milk, incubated with anti- α -tubulin antibody (1:2000) and anti-NF-L antibody (1:2000) separately overnight at 4 °C, and then incubated with HRP-conjugated anti-rabbit secondary antibody (1:8000) for 1 h at room temperature. An ECL Plus Western Blotting Substrate Kit (Thermo Fisher Scientific, USA) was used to visualize the immunoreactivity on X-ray film. X-ray films were scanned by a V3 Western Workflow Complete System for Mini Gels. Protein bands were quantified with Gel-Pro Analyzer software.

β -integrin expression in the brainstem of the deceased that died from TAI

The paraffin section of the brainstem of the deceased that died from TAI and from nonviolent cause (coronary heart disease)

was obtained by routine histology. Immunohistochemistry of β -integrin expression in the brainstem was referred to as β -APP expression in TAI rats.

Statistical analysis

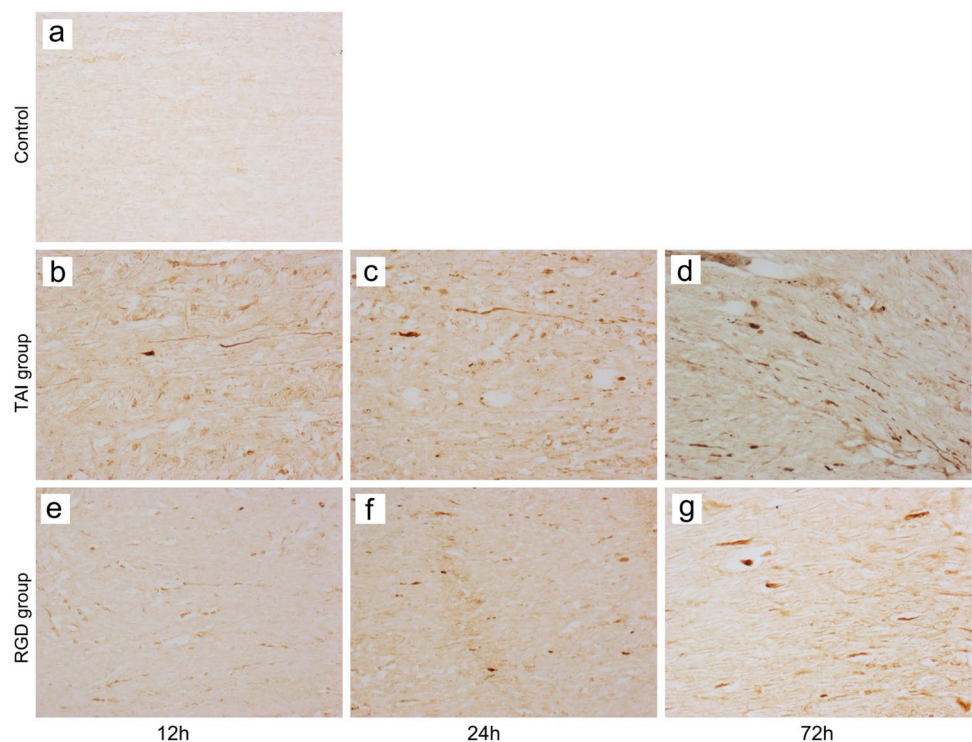
The data are shown as the mean $\pm$ SEM. For all the experimental data, we used Prism 5.0 (GraphPad Software Inc., 108 La Jolla, CA, USA) to perform a one-way ANOVA. $P < 0.05$ was considered statistically significant.

Results

Integrin affects β -APP expression in the brainstem of TAI rats

β -APP immunohistochemistry was used to observe axonal injury. The results showed that the injured axon was β -APP-positive. The swelling and distortion of the axon can be seen 12 h postinjury, while axonal varicosities and axonal retraction balls (ARBs) were distinct at 24 h and markedly increased at 72 h postinjury. The intensity of β -APP expression is increased with time after trauma. However, no axon injuries were observed in the control group, and no β -APP-positive axons were observed in the control group. After RGD administration, β -APP expression was relatively weaker than that in the TAI group (Fig. 1).

Fig. 1 β -APP expression in rat brainstem (IHC $\times 200$). **a** The control group did not show expression of β -APP, **b–d** the TAI group showed strong expression of β -APP, **e–g** the RGD group showed expression of β -APP but at lower levels than in the TAI group



Integrin affects microtubules and neurofilaments in the brainstem of TAI rats

According to our previous study [24], the ultrastructural changes of TAI were most severe at 24 h postinjury, and that time interval was chosen to observe the ultrastructure of microtubules and neurofilaments in myelinated and unmyelinated axons. The TEM results showed that the lesions of unmyelinated fibers were more significant than those of myelinated fibers, and the microtubules and the neurofilaments showed misalignment, disintegration, and loss. RGD could effectively alleviate the injury of the microtubules and microfilaments (Fig. 2).

Immunoblot analysis of axonal α -tubulin and NF-L alterations

A microtubule is a heterodimer composed of alpha and beta tubulin. Our study showed that α -tubulin was expressed abundantly in neuron soma and axons of the brainstem in control and TAI rats (immunohistochemistry results not shown). Immunoblot analysis of α -tubulin in brainstems from TAI rats revealed significant protein loss at all three-time points postinjury relative to controls ($P < 0.05$). Among the time points we observed, the posttraumatic α -tubulin

loss was most serious at 24 h, and then a small increase was detected at 72 h. In the RGD group, the α -tubulin level showed a significant increase at 24 h and 72 h compared to the TAI group ($P < 0.05$) and even reached the level of control rats at 72 h (Fig. 3).

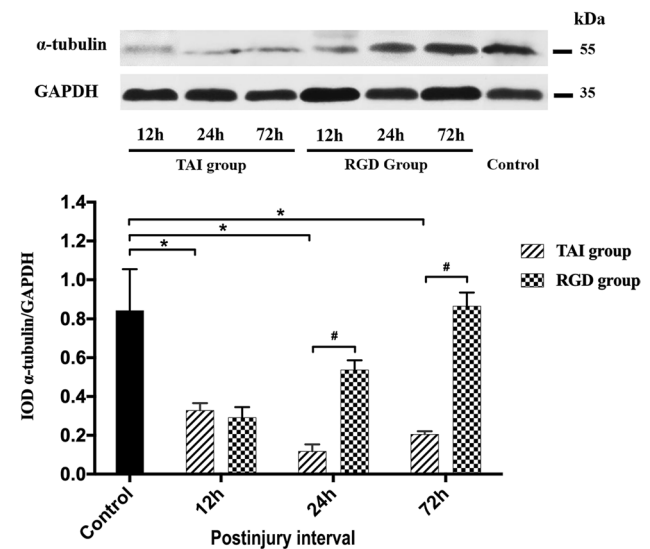
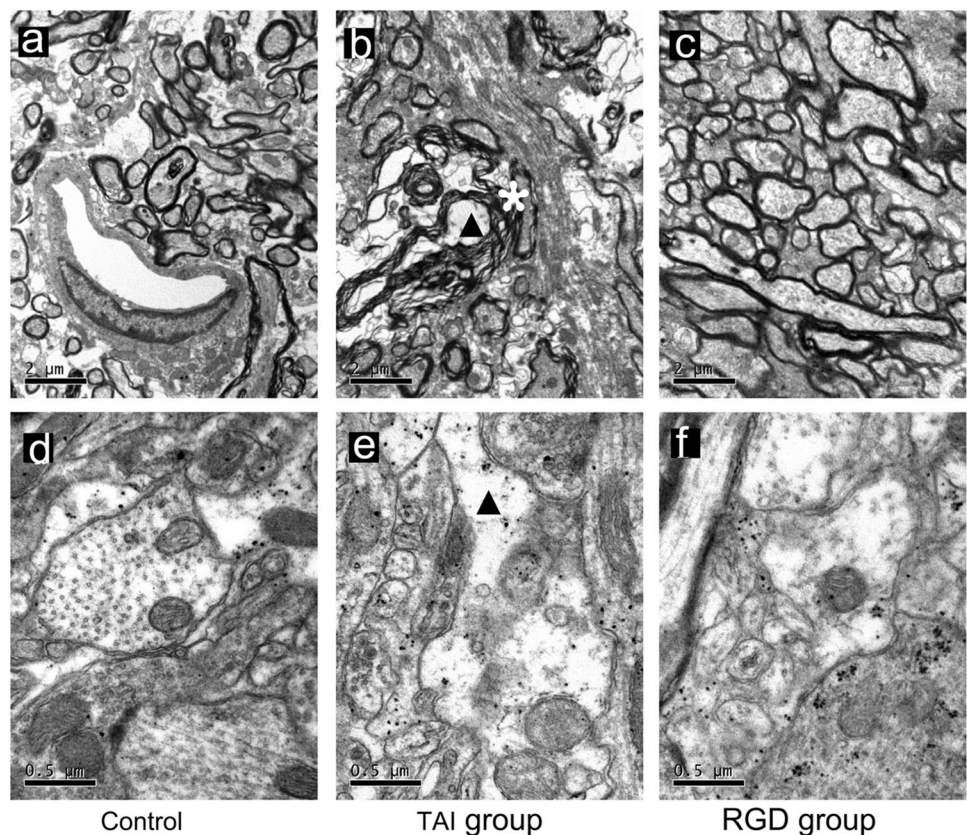


Fig. 3 Immunoblot analysis of α -tubulin expression in rat brainstem. α -Tubulin expression in brainstems was lower in the TAI group than in the controls and was higher in the RGD group than in the TAI group

Fig. 2 The ultrastructure of microtubules and microfilaments in rat brainstem at 24 h postinjury. **a–c** Myelinated nerve fibers and myelin lamellar structures were severely separated in some axons (asterisk). The microtubules and microfilaments were disorganized, disintegrated, and even completely absent (triangle), and only residual axolemma can be observed in some axons. **d–f** Unmyelinated nerve fibers showed more severe cytoskeletal damage (triangle). The extent of axonal injury in the RGD group was significantly milder than in the TAI group



A neurofilament is made up of at least two of the three protein subunits, neurofilament light (NF-L), neurofilament medium (NF-M), and neurofilament heavy (NF-H), wherein NF-L is an essential ingredient. When neurofilaments are injured, the subunits of neurofilaments separate; thus, NF-L expression is increased in TAI. In the present study, NF-L presented extremely weak staining in the brainstem of control rats. The immunohistochemistry profiles, including axonal swellings, and axonal retraction balls, were clearly NF-L-positive in the brainstem of the TAI rats (immunohistochemistry results not shown). Immunoblot analysis showed that the most significant increase in NF-L levels was at 12 h in TAI. The NF-L level was decreased at 24 h and 72 h in TAI but was still higher than that in the control group ($P < 0.05$). The administration of RGD significantly decreased NF-L expression from 12 h ($P < 0.05$) to 24 h ($P < 0.05$) postinjury compared with TAI (Fig. 4).

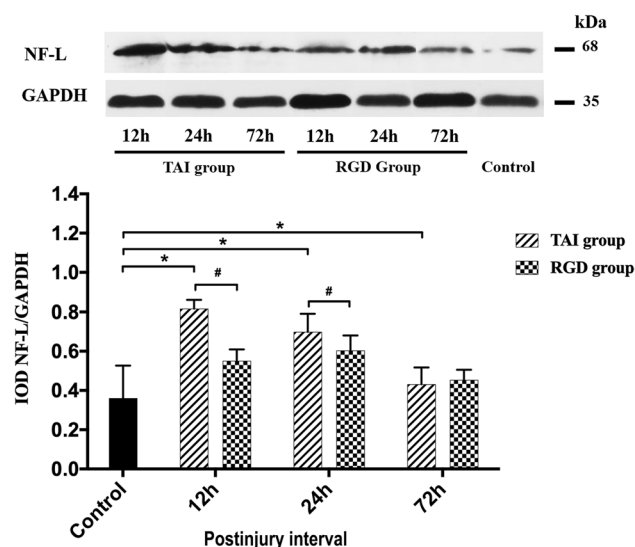
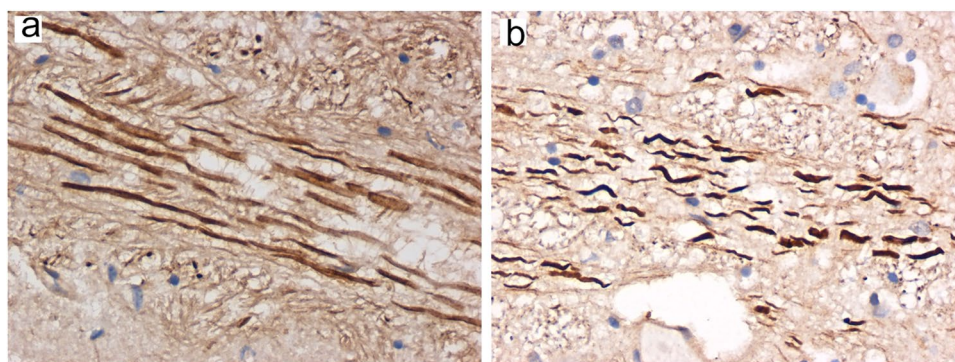


Fig. 4 Immunoblot analysis of NF-L expression in rat brainstem. NF-L expression was increased in TAI and was lower in the RGD group than in the TAI group

Fig. 5 The expression of β -integrin in the pons of forensic cases. **a** The pons of an individual who died from coronary heart disease showed expression of β 1-integrin. **b** The pons of an individual who died from TAI showed stronger, more centralized β 1-integrin expression



Integrin expression in the brainstem of the deceased that died from TAI

Finally, we observed the expression of β -integrin in the brainstem of the deceased that died from TAI, which showed that β -integrin can be expressed in the normal brainstem, but the expression tended to be centralized and the intensity was obviously stronger in the brainstem of the deceased from TAI compared with the deceased from nonviolent death (Fig. 5).

Discussion

TAI usually results from head trauma involving fast acceleration and/or deceleration, which is commonly seen in motor vehicle accidents, falls from height, and blunt impacts [25]. Sudden head movement produces a force vector inside the intracranial cavity, which can lead to shear and tearing of axons [26]. Histopathological findings in TAI have a predisposition in the white matter tracts in the midline of the brain, including the grey–white interface, the corpus callosum, and brainstem [27–29]. In our previous study, we observed diffuse axonal injury in the corpus callosum and brainstem in the Marmarou rat model [22]. Marmarou model has the advantages of low cost, simplicity, and the ability to replicate different degrees of damage easily. Here, we used Marmarou model to observe the mechanism of TAI, and the brainstem was selected as the region of interest considering the important contribution of brainstem injury to coma and death.

The pathophysiological mechanism of TAI is still undefined, although it is accepted that it is due to secondary damage. Damage to the cytoskeleton, especially microtubule impairment, is thought to be a key element of axon disconnection [30]. The injuries of microtubules and neurofilaments in the axon are not only a response to trauma but also cause a series of subsequent biochemical reactions in TAI [31]. The microtubules and microfilaments in animal models and humans with TAI have also been described

by many researchers [30, 32, 33]. It has been reported that within 72 h post TAI, α -tubulin expression was significantly decreased [34]. Another study showed a focal increase in NF-L-positive staining 30 min after mild TBI [35, 36]. In addition, Lei et al. [37] found that β 1-integrin signalling plays an instructive role in axon initiation and growth by regulating microtubule assembly. In the present study, with the combination of ultrastructure observation and cytoskeleton protein quantification, we studied the effect of integrin on the microtubules and neurofilaments of axons in TAI. We also observed β -integrin immunostaining in the brainstem in the forensic case, which indicated a difference in β -integrin expression between TAI-induced death and nonviolent death. Our study indicated that integrin-mediated intracellular signalling contributes to microtubule and neurofilament injury in axonal damage caused by trauma.

Histopathology assessment of TAI was only applicable in postmortem diagnosis. However, TAI, as a characteristic white matter pathology, is difficult to detect because of subtle and indistinct morphological abnormalities. Although β -APP is thought to be a more sensitive marker for the diagnosis of TAI than HE and silver staining, its application in the differential diagnosis between traumatic and nontraumatic axonal injury is still being studied [38]. The specificity of β -APP immunohistochemistry in TAI diagnosis has yet to be recognized. Hence, it is imperative to diagnose TAI by the combination of multiple biomarkers that could reflect the mechanisms of TAI. Through our research, we hope to provide new insights into the biomechanical mechanism of TAI and potential useful biomarkers for TAI. The findings of this study suggest that integrin signal-mediated cytoskeletal injury plays a role in the pathogenic mechanism of TAI and that β -integrin has the potential to be used as a postmortem diagnostic indicator of TAI. However, the differences in biomechanical mechanisms between simple animal models and highly complex human TAI should be considered, and further study is needed.

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Data availability The data that support the findings of this study can be freely available to any researcher wishing to use them for non-commercial purposes.

Declarations

Ethics approval The experiments in this study were conducted in accordance with the guidelines of Huazhong University of Science and Technology for the care and use of animals.

Conflict of interest The authors declare no competing interests.

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